Exhaustiveness of tableaux derivability

The Open Logic Project contributors

openlogic:first-order-logic/tableaux:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-tableaux/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source's phrase 'On the left left side' is the author's own typo and is delivered verbatim; no tableau environment occurs in the counted span

Editorial note. The label prop:provability-exhaustive names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Inconsistency via closed tableaux. A finite signed assumption set is inconsistent iff it has a closed tableau; inconsistency of a set is witnessed by a finite subset with a closed tableau.. Both antecedent inconsistencies are unpacked into closed-tableau witnesses with True A (resp. True not-A) among the assumptions.
- The Cut rule. A tableau branch may be split on any sentence A into a branch with True A and one with False A.. The single Cut application routes each half of the combined tableau to its own closing witness.

Proposition 0.1. *If* $\Gamma \cup \{A\}$ *and* $\Gamma \cup \{\neg A\}$ *are both inconsistent, then* Γ *is inconsistent.*

Proof. If there are B_1, ..., $B_n \in \Gamma$ and C_1, ..., $C_m \in \Gamma$ such that

$$\{\top(A), \top(B_1), \ldots, \top(B_n)\} \text{ and}$$
$$\{\top(\neg A), \top(C_1), \ldots, \top(C_m)\}$$

both have closed tableaus, we can construct a single, combined tableau that shows that Γ is inconsistent by using as assumptions $\top(B_1)$, ..., $\top(B_n)$ together with $\top(C_1)$, ..., $\top(C_m)$, followed by an application of the Cut rule. This yields two branches, one starting with $\top(A)$, the other with $\bot(A)$.

On the left left side, add the part of the first tableau below its assumptions. Here, every rule application is still correct, since each of the assumptions of the first tableau, including $\top(A)$, is available. Thus, every branch below $\top(A)$ closes.

On the right side, add the part of the second tableau below its assumption, with the results of any applications of $\neg$ to $\top(\neg A)$ removed. The conclusion of $\neg$ to $\top(\neg A)$ is $\bot(A)$, which is nevertheless available, as it is the conclusion of the Cut rule on the right side of the combined tableau.

If a branch in the second tableau was closed because it contained the assumption $\top(\neg A)$ (which no longer appears as an assumption in the combined tableau) as well as $\bot(\neg A)$, we can applying $\neg$ to $\bot(\neg A)$ to obtain $\top(A)$. Now the corresponding branch in the combined tableau also closes, because it contains the right-hand conclusion of the Cut rule, $\bot(A)$. If a branch in the second tableau closed for any other reason, the corresponding branch in the combined tableau also closes, since any signed formulas other than $\top(\neg A)$ occurring on the branch in the old, second tableau also occur on the corresponding branch in the combined tableau. $\square$